FY-926

FIRST YEAR HIGHER SECONDARY
IMPROVEMENT/SUPPLEMENTARY EXAMINATION, OCTOBER 2026

Part – III

BIOLOGY
(Botany & Zoology)

Maximum : 60 Scores

Time : 2 Hours
Cool-off time : 15 Minutes
Preparatory Time : 10 Minutes

General Instructions to Candidates :

- There is a 'Cool-off time' of 15 minutes in addition to the writing time. Further there is a '10 minutes' 'Preparatory Time' at the end of the Botany Examination and before the commencement of Zoology Examination.
- Use the 'Cool-off time' to get familiar with questions and to plan your answers.
- Read questions carefully before answering.
- Read the instructions carefully.
- Calculations, figures and graphs should be shown in the answer sheet itself.
- Malayalam version of the questions is also provided.
- Give equations wherever necessary.
- Electronic devices except non-programmable calculators are not allowed in the Examination Hall.

വിദ്യാർത്ഥികൾക്കുള്ള പൊതുനിർദ്ദേശങ്ങൾ :

- നിർദ്ദിഷ്ട സമയത്തിന് പുറമെ 15 മിനിറ്റ് 'കൂൾ ഓഫ് ടൈം' ഉണ്ടായിരിക്കും. കൂടാതെ ബോട്ടണി പരീക്ഷയ്ക്കുശേഷം സുവോളജി പരീക്ഷ തുടങ്ങുന്നതിനുമുമ്പ് '10 മിനിറ്റ്' തയ്യാറെടുപ്പുകൾ നടത്തുന്നതിനായി നല്കുന്നതാണ്.
- ചോദ്യങ്ങൾ പരിചയപ്പെടാനും ഉത്തരങ്ങൾ ആസൂത്രണം ചെയ്യാനും 'കൂൾ ഓഫ് ടൈം' ഉപയോഗിക്കുക.
- ഉത്തരങ്ങൾ എഴുതുന്നതിനു മുമ്പ് ചോദ്യങ്ങൾ ശ്രദ്ധാപൂർവം വായിക്കണം.
- നിർദ്ദേശങ്ങൾ മുഴുവനും ശ്രദ്ധാപൂർവം വായിക്കണം.
- കണക്ക് കൂട്ടലുകൾ, ചിത്രങ്ങൾ, ഗ്രാഫുകൾ, എന്നിവ ഉത്തരപേപ്പറിൽ തന്നെ ഉണ്ടായിരിക്കണം.
- ചോദ്യങ്ങൾ മലയാളത്തിലും നല്കിയിട്ടുണ്ട്.
- ആവശ്യമുള്ള സ്ഥലത്ത് സമവാക്യങ്ങൾ കൊടുക്കണം.
- പ്രോഗ്രാമുകൾ ചെയ്യാനാകാത്ത കാൽക്കുലേറ്ററുകൾ ഒഴികെയുള്ള ഒരു ഇലക്ട്രോണിക് ഉപകരണവും പരീക്ഷാഹാളിൽ ഉപയോഗിക്കുവാൻ പാടില്ല.

PART – A

BOTANY

(Maximum : 30 Scores) **Time : 1 Hour**

I. Answer any 3 questions from 1 to 5. Each question carries 1 score. **(3 × 1 = 3)**

1. Nitrogen fixing cells seen in Anabaena is called _____ **Answer : Heterocyst**

2. Which of these algae produce the hydrocolloidal substance called carrageen ?

   (a) Green algae (b) Brown algae

   (c) Red algae ✓ (d) All of these

3. If a fruit is formed without fertilisation of the ovary, it is called a _____ fruit.

   Parthenocarpic

4. The process of division of nucleus during cell division is known as _____. **Karyokinesis**

5. Which is the first isolated naturally existing cytokinin ? **Zeatin**

II. Answer any 9 questions from 6 to 16. Each question carries 2 scores. **(9 × 2 = 18)**

6. (a) Which are the two substances formed as result of breakdown of fat during respiration ?

   (b) It would be better to consider the respiratory pathway as an amphibolic pathway rather than as a catabolic one. Why ?

(a) Fatty acids, Glycerol

(b) Because respiration involves both catabolism and anabolism .

7. The life cycle of Bryophytes has two stages.

(a) Which are they ?

(b) Write one difference between them.

(a) Gametophyte, Sporophyte

(b)

- **Gametophyte : It is the dominant, independent, and photosynthetic phase that produces gametes.**
- **Sporophyte: It is short-lived, dependent on the gametophyte for nutrition, and produces spores.**

8. By observing the functions of plant hormones given in the first column, write the name of plant hormone that is responsible for this function in second column.

Function of plant hormone	Name of plant hormone
(a) Ripening of fruits	(i) .....
(b) Bolting	(ii) .....
(c) Inducing seed dormancy	(iii) .....
(d) Apical dominance	(iv) .....

(i) Ethylene
(ii) Gibberellin
(iii) Abscisic acid
(iv) Auxin

9. (a) Name the class of fungi commonly known as sac fungi.

(b) Which fungus of this class is extensively used for biochemical and genetical studies ?

(a) Ascomycetes

(b) Neurospora

10. Write one difference between the following :

(a) Open and Closed vascular bundle

(b) Conjoint and Radial vascular bundle

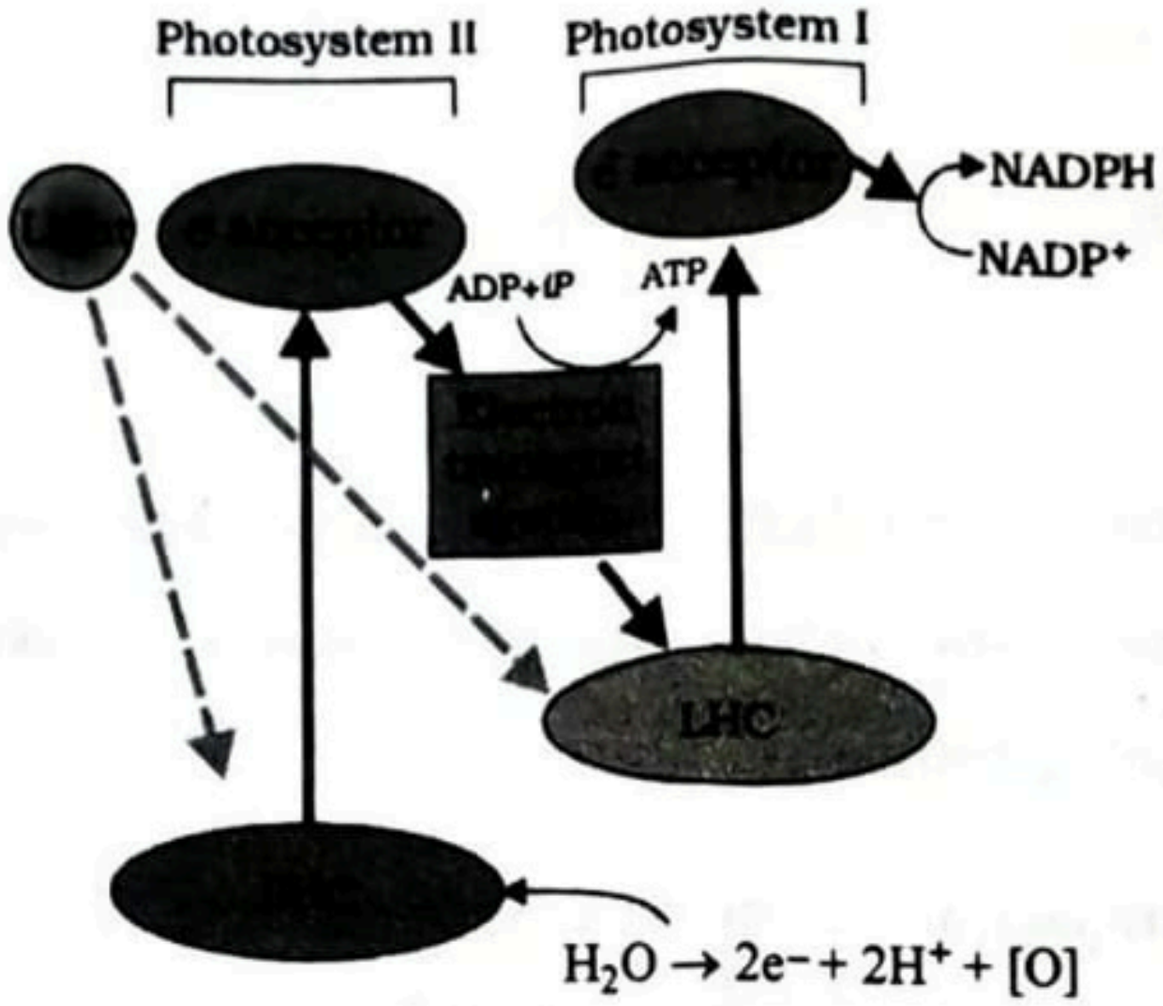
(a)

Open VB	Closed VB
Contains cambium between xylem and phloem, allowing for secondary growth (found in dicots).	Lacks cambium, so no secondary growth occurs (found in monocots).

(b)

Conjoint vascular bundle	Radial vascular bundle
Xylem and phloem are arranged along the same radius within the bundle (found in stems and leaves).	Xylem and phloem are arranged alternately on different radii (found in roots).

11. Observe the given figure and answer the following questions :



(a) Name the process.

(b) Where does it take place in Chloroplast ?

(c) What is the significance of splitting of water in this process ?

(a) Non cyclic photophosphorylation
(b) Thylakoid membrane (or Grana lamellae).
(c) It provides electrons to continuously replace those lost by Photosystem II (PS II), and releases protons oxygen.

12. (a) Write the name of any two cell organelles bounded by two membranes.

(b) Write one function of each.

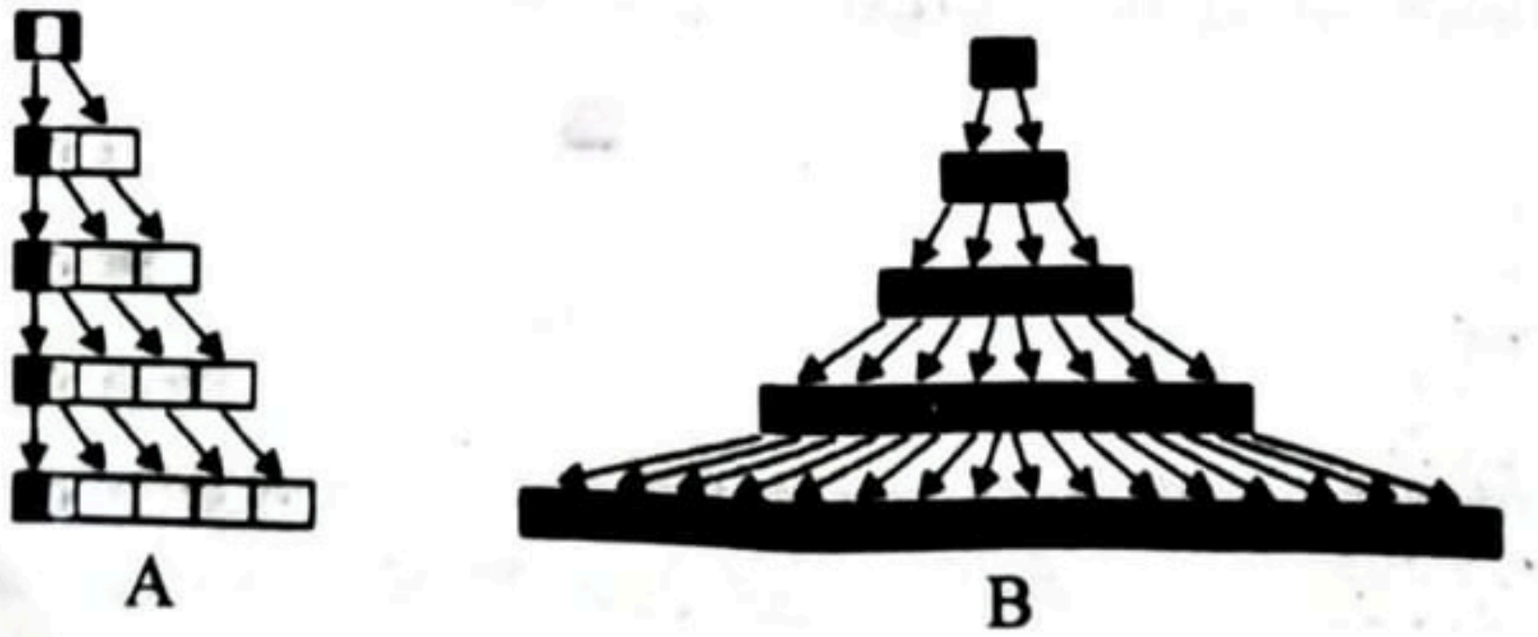
(a) 1. Mitochondria
2. Chloroplast (Nucleus)
(b)

- **Mitochondria: Cellular respiration and ATP production (powerhouse of the cell).**
- **Chloroplast: Performs photosynthesis to synthesize food/carbohydrates using sunlight.**

13. Observe the figures A and B and answer the following questions :

(a) Which are the two types of growth rates represented in figure A and B ?

(b) Write the difference between these two growth rates.



(a) Figure A: Arithmetic growth
Figure B: Geometric growth

(b)

Arithmetic growth (A)	Geometric growth (B)
Following mitotic cell division, only one daughter cell continues to divide, while the other differentiates and matures.	Following mitotic cell division, both daughter cells retain the ability to divide and continue to do so.

14. In grasses, how Bulliform cells helps to minimise water loss from leaves ?

When water is scarce, bulliform cells lose water and become flaccid. This causes the leaves to curl inwards, reducing the exposed surface area and minimizing further water loss through transpiration

15. (a) Why C4 plants have better productivity and yield than C3 plants ?

(b) Write two examples of C4 plants.

(a) C4 plants have better productivity and yield because they have negligible photorespiration and a more efficient CO_2 fixation mechanism.

(b) Maize and sugarcane.

16. (a) The cell membrane is mainly consists of two substances. Which are they ?

(b) Write the difference between Passive transport and Active transport.

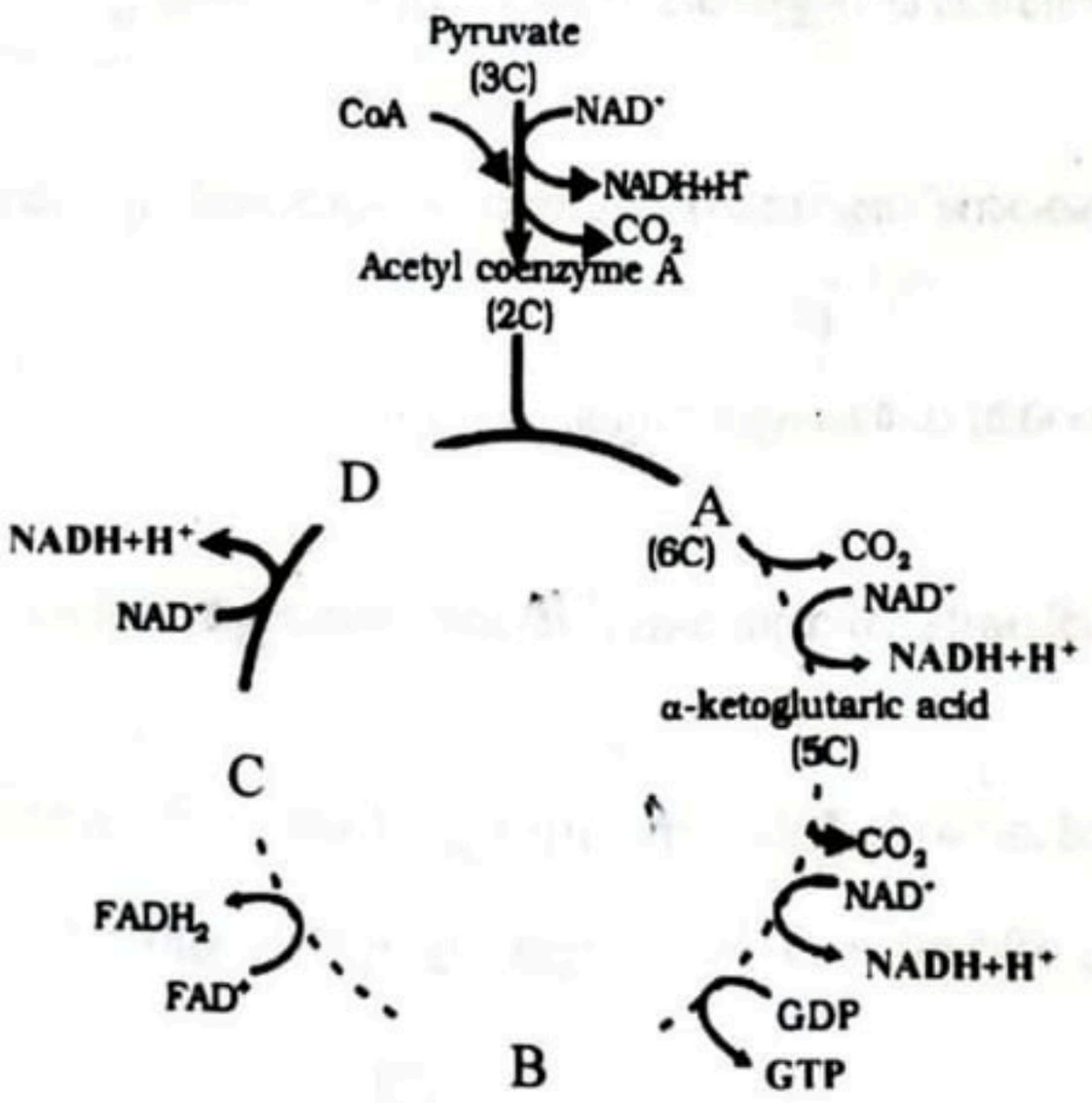
(a) Lipids and proteins

(b)

- **Passive transport: Substances move along the concentration gradient without energy.**
- **Active transport: Substances move against the concentration gradient using energy (ATP).**

III. Write any 3 questions from 17 to 20. Each question carries 3 scores. **(3 × 3 = 9)**

17. A process of aerobic respiration is given in the picture. Observe the picture and answer the following questions :



(a) Name the process.

(b) Write the names of compounds marked as A,B,C and D in the picture.

(c) Where does this process occur in the cell ?

(a) Kreb's cycle

(b) A- Citric acid B- Succinic acid

C- malic acid D- Oxaloacetic acid

(c) Mitochondrial matrix

18. Briefly describe the following terms associated with monocotyledonous seed :

(a) Aleurone layer

(b) Scutellum

(c) Coleorhiza

(a) Aleurone layer: The protein-rich outer layer of the endosperm in monocot seeds that surrounds the storage tissue and secretes digestive enzymes during germination.

(b) Scutellum: The single cotyledon of a monocot seed.

(c) Coleorhiza: Protective sheath covering the radicle in monocot seeds.

19. Write three differences between C3 plants and C4 plants.

C3 Plants	C4 Plants
First stable product has 3 carbon atoms.	First stable product has 4 carbon atoms.
Photorespiration is present.	Photorespiration is negligible.
Kranz anatomy is absent.	Kranz anatomy is present.

20. Some of the events of the stages of Mitosis is given below. Write them in their appropriate columns of the table :

(i) Spindle fibres attach to kinetochores of chromosomes

(ii) Nuclear envelope develops around the chromosome clusters at each pole forming two daughter nuclei

(iii) Centromeres split and chromatids separate

(iv) Chromosomal material condenses to form compact mitotic chromosomes

(v) Chromatids move to opposite poles

(vi) Centrosome which had undergone duplication during interphase, begins to move towards opposite poles of the cell

Pro phase	Meta phase	Ana phase	Telo phase

Prophase	Metaphase	Anaphase	Telophase
(iv),(vi)	(i)	(iii), (v)	(ii)